



Learning from animals

A drone with a sonar array gets its inspiration from a bat. Check it out here:
<http://dgit.in/1AgJCKm>

Lighting up the way

Philips has introduced a new indoors navigation system featuring Visual Light Communication and LEDs:
<http://dgit.in/1HAGvDg>

DO



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Airflow 101

Be it an entry-level or a high-end rig, keeping it cool is top priority. The first step towards maintaining proper component temperature is optimising airflow.



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There are a host of utilities that can help monitor components such as the processor and GPU in addition to optimising settings to maintain optimum temperatures. However, it's easy to forget that ambient temperatures can also affect component integrity as well as performance. While enthusiasts go for elaborate liquid cooling systems, properly managed airflow is half the battle won. What follows are some basic rules to

keep in mind while designing an airflow management system for your rig. Before we begin, it bodes well to know that airflow is at its optimum if intake and outflow are in line of sight.

1. COLD AIR IN, HOT AIR OUT

The basic rule is to keep the air inside the case moving. If the air within the system enclosure is static, the ambient temperature will rise and adversely affect performance. This is often a consequence of a significantly imbalanced airflow design. The rate of intake and outflow can be attributed to a number of factors such as size, airflow, and speed of a fan. Even factors such as the size of perforations for ventilation affect airflow. To ensure lower ambient temperatures, always verify the direction of airflow before securing any fan to the system case. If a fan is behind a grille, make sure the perforations are large enough to provide sufficient airflow. Removing the system grille occluding the fan is a great way to reduce noise and increase airflow. Bigger the opening, better the cooling.

2. HOT AIR RISES

Hot air is less dense and tends to rise upward. This phenomenon makes the flow of air inside the case to move in a lazy 'Z' manner. The cold intake is from the front lower side of the case whereas the hot air escapes from the rear top side of the case. This setup is ideal, unless extra fans are employed for cooling hot-spots. The side fans, especially, create a whirlpool effect and interfere with the 'Z' pattern, leaving dead spots in the case. If spot cooling is required, for the processor or GPU, use air ducts that direct the side fan to the target.

3. CHOOSE THE RIGHT FAN FOR YOUR RIG

There are three kinds of fans: airflow optimised, static pressure optimised and, as always, the hybrid. An airflow optimised



fan has a relatively bigger gap between its blades to facilitate air movement. This is useful when the fan isn't addled with resistance from any components such as grilles or radiators. If the fan pushes air against a radiator, the air will take the path of least resistance and flow around it.

This is where static pressure optimised fan come into the picture. Such fans leaves almost no room between the blades to form a suction and force more air to flow through it. It also has a wide exhaust angle, and is hence pressure optimised. The hybrid fan strikes a balanced between these two designs and is characterised by a blade gap that lies between

the two designs. Hybrid designs may sometimes feature blades of differing shapes in a single fan.

4. PAY ATTENTION TO RATINGS

Look for fans that have high airflow rating. The rating is usually expressed in CFM (Cubic Feet per Minute) and denotes the maximum volume of air that the fan can move in one minute. Higher the airflow value, fewer the fans you need. A fan with high static pressure rating, which is measured in mm/H₂O, works the best when used in conjunction with more obstructive applications such as radiators.

5. KNOW YOUR AIRFLOW SYSTEM

Using more fans doesn't always translate into better cooling performance. Sub-optimal positions and choosing the wrong fan type can lower cooling efficiency. There are three primary types of airflow systems that you can consider.

NEGATIVE PRESSURE – LESS IN, MORE OUT

In this type of airflow setup exhaust fans outnumber intake fans. Since there's less intake, the system relies more on ventilation for air inflow. This is why negative systems are optimal when